

BHU-NETRA: Satellite Ground Segment & Mission Control Platform

Document 03: Dual Architecture Design – Open Source vs. Enterprise Proprietary

1. Introduction & Comparative Philosophy

To satisfy national strategic autonomy, sovereign security, and technology evaluation guidelines, two complete end-to-end architectures have been designed for **BHU-NETRA**:

1. **Architecture Stack A: Indigenous & Open-Source First Stack**

- *Philosophy*: Built on proven open-source frameworks, open standards (OGC, CCSDS, VITA 49), indigenous libraries, and open-source geospatial tools. Eliminates vendor lock-in, allows full source-code auditability, and provides complete strategic autonomy for air-gapped defense and government installations.

2. **Architecture Stack B: Non-Open Source / Enterprise Proprietary Stack**

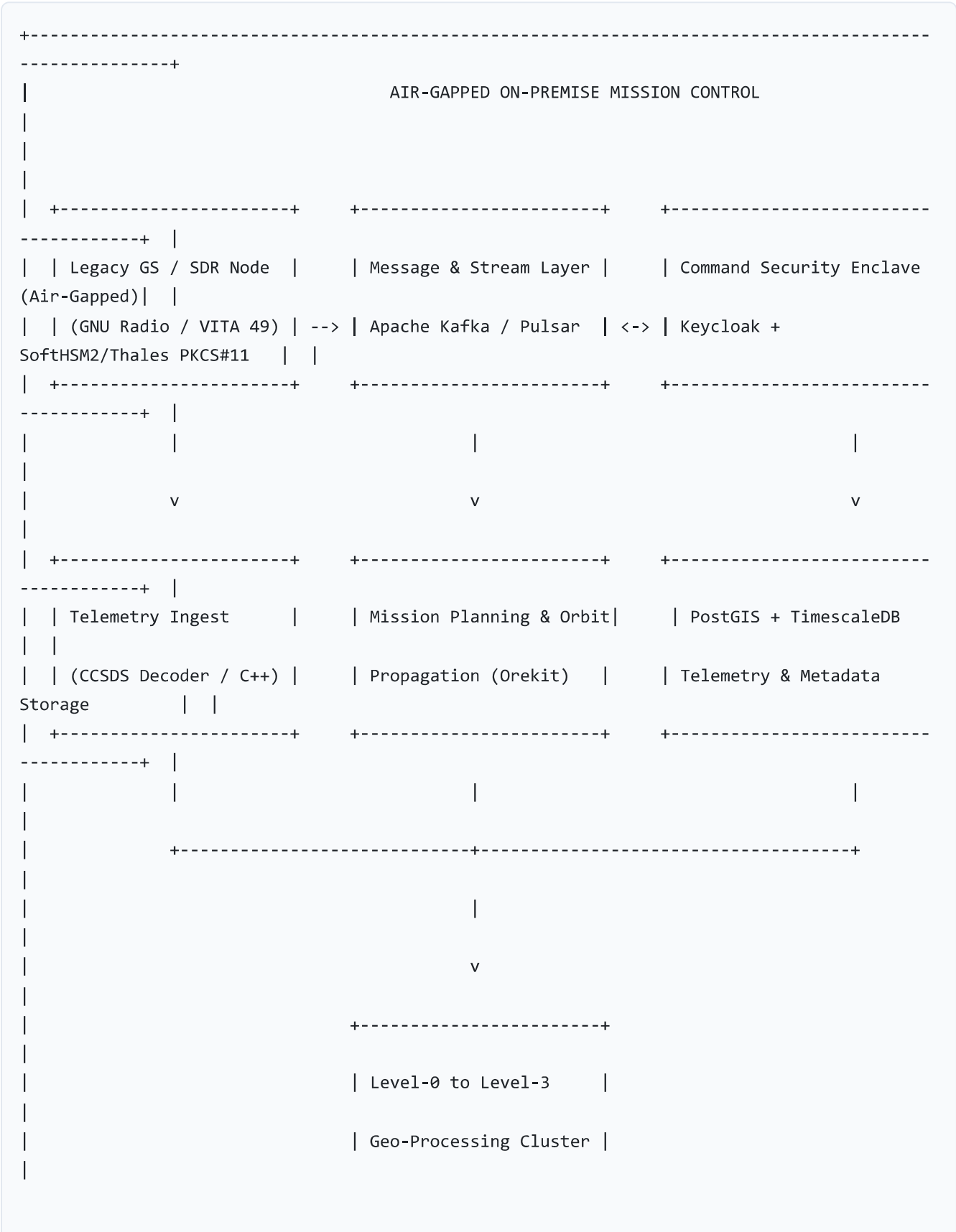
- *Philosophy*: Built on turn-key commercial enterprise suites (Ansys STK, Oracle, IBM MQ, ESRI ArcGIS, Kratos Quantum). Offers commercial SLAs, out-of-the-box regulatory compliance, and off-the-shelf support contracts at higher licensing costs.

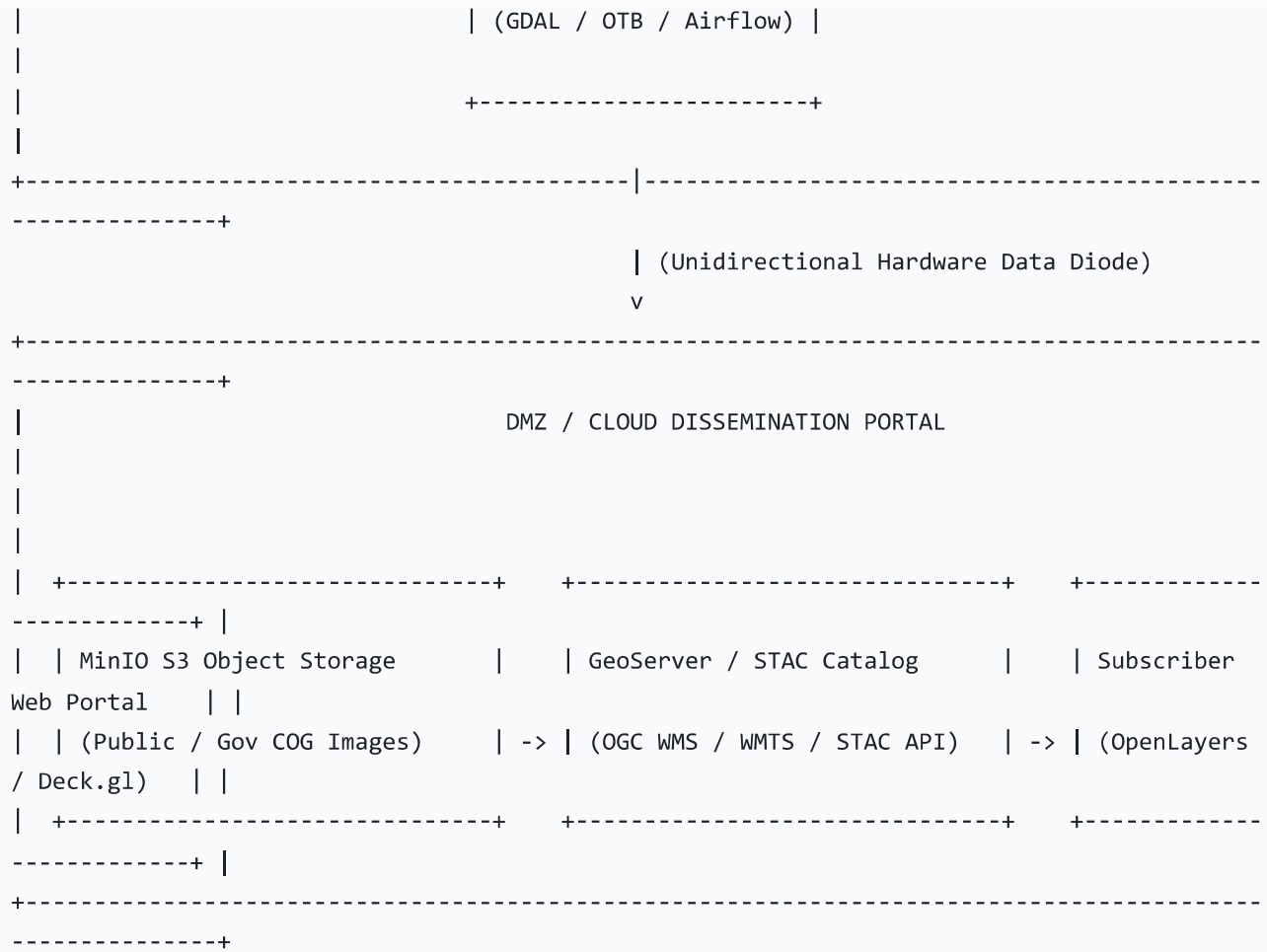
2. Side-by-Side Architectural Layer Comparison

Architectural Layer	Architecture Stack A (Indigenous / Open-Source)	Architecture Stack B (Enterprise / Proprietary)
Operating System	Rocky Linux 9 / Red Hat Enterprise Linux (Community Build) / Ubuntu 22.04 LTS (Kernel hardened)	Red Hat Enterprise Linux (RHEL) Enterprise Server with Red Hat Satellite)
Orbital Dynamics & Propagation	Orekit (Java) + GDT (Ground Segment Development Toolkit) + Python SGP4	Ansys STK (Satellite Tool Kit) Systems Tool Kit Engine + Orbit Determination Tool Kit (ODTK)
Baseband & SDR Processing	GNU Radio 3.10 + VITA 49 (Libvita49) + USRP UHD API	Kratos Quantum Radio / AWS Ground Station API / RT Logic Telemetry Processors
Message Broker & Stream Ingest	Apache Kafka (Core Event Stream) + Apache Pulsar / EMQX (Telemetry MQTT)	IBM MQ Enterprise + Solace PubSub+ Event Broker
Relational & Spatial Database	PostgreSQL 16 + PostGIS 3.4	Oracle Database 19c Enterprise Edition + Oracle Spatial and Graph
Time-Series Telemetry DB	TimescaleDB (PostgreSQL Extension) / Apache Druid	InfluxDB Enterprise / AVEVA PI System (OSIsoft PI)
Object Storage (Payload)	MinIO Enterprise (S3 Compliant, Multi-Node Erasure Coding) / Ceph	NetApp StorageGRID / Dell EMC Elastic Cloud Storage (ECS)
Geospatial Processing Engine	GDAL / OGR + Orfeo ToolBox (OTB) + PDAL + Python Rasterio	ESRI ArcGIS Enterprise Server + ERDAS IMAGINE Engine
Map Rendering & OGC Services	GeoServer 2.24 + OpenLayers 8 / Deck.gl	ESRI ArcGIS Image Server + ArcGIS Portal Map Viewer
Orchestration & Workflow	Apache Airflow 2.8 + Kubernetes / K3s (Air-gapped cluster)	VMware Tanzu Application Platform / Red Hat OpenShift Container Platform
Identity & Access Control	Keycloak 23 (OpenID Connect / OAuth2 / SAML2) + FreeIPA (LDAP)	Ping Identity Enterprise + Microsoft Active Directory / CyberArk
Hardware Security Module (HSM)	Thales Luna HSM (PKCS#11 Open API) / SoftHSM2 (Dev)	Thales PayShield / SafeNet Network HSM Enterprise Suite
Audit & Logging	OpenSearch 2.11 / Vector	Splunk Enterprise + IBM QRadar

3. Detailed Architecture Stack A: Indigenous & Open-Source First Stack

3.1 Core Architecture Diagram (Open-Source Stack)



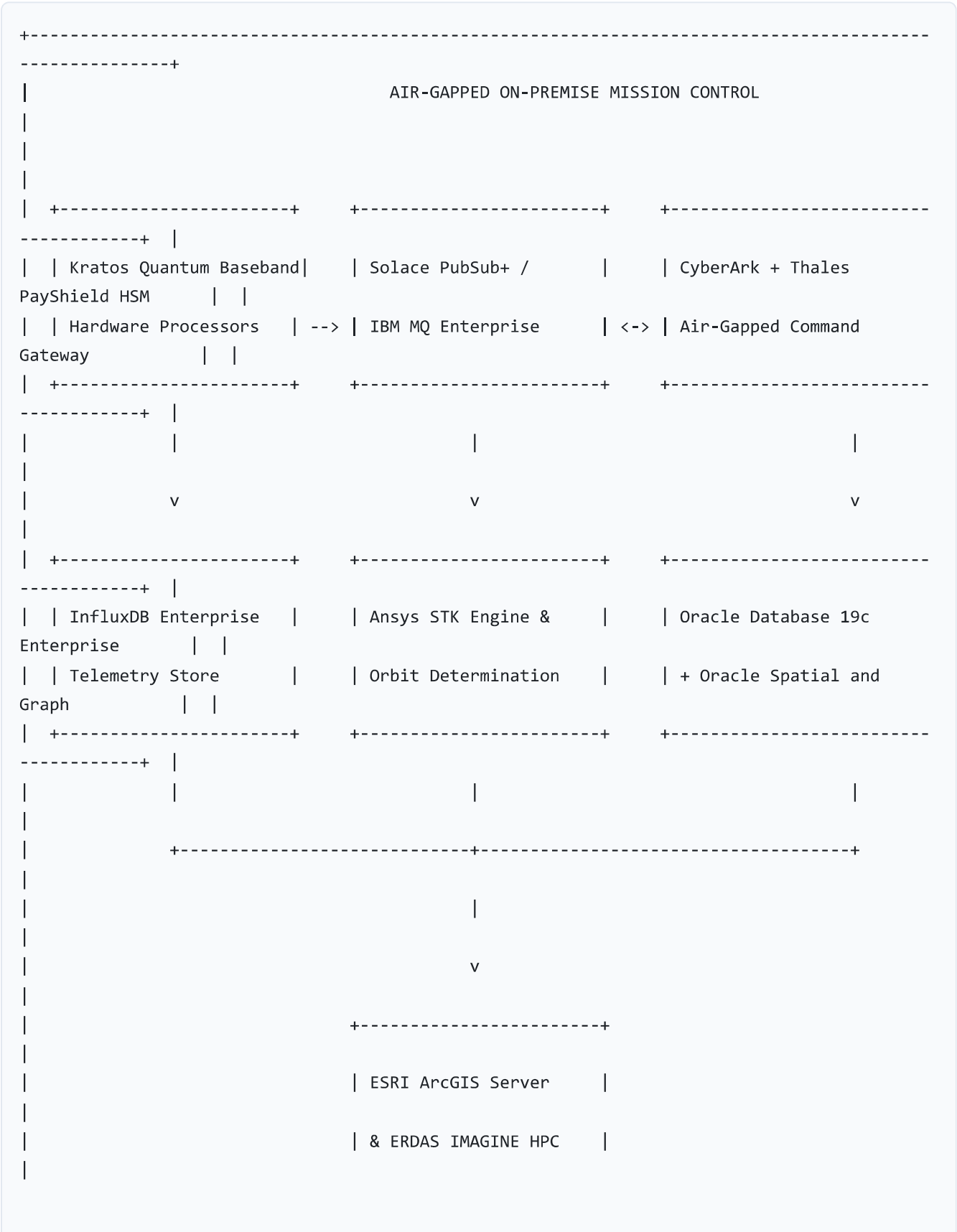


3.2 Strengths of Open-Source Architecture Stack A

1. **Complete Sovereignty & Security Auditability:** Full access to source code ensures zero backdoors and adherence to strict defense air-gap policies.
2. **Zero Recurring License Fees:** Scalable to 100+ satellites and 20+ ground stations without per-core or per-satellite licensing costs.
3. **High Adaptability:** Customized DSP algorithms can be added directly into GNU Radio or Orekit.

4. Detailed Architecture Stack B: Enterprise Proprietary Stack

4.1 Core Architecture Diagram (Proprietary Stack)





4.2 Strengths of Proprietary Architecture Stack B

1. **Commercial Vendor SLAs:** Turn-key support from Ansys, Oracle, ESRI, and IBM with guaranteed incident response times.
2. **Pre-Certified Compliance:** Out-of-the-box compliance with international space standards and enterprise IT frameworks.
3. **Turn-Key GUIs:** Rich GUI applications for STK flight dynamics officers and ArcGIS GIS analysts.

5. Architectural Evaluation Matrix

Evaluation Criteria	Stack A (Open-Source / Indigenous)	Stack B (Enterprise Proprietary)	Recommended Winner
Strategic Autonomy	10 / 10 (Full control over code & algorithms)	3 / 10 (Subject to export controls / sanctions)	Stack A
Air-Gap Feasibility	10 / 10 (Zero phone-home / offline license keys)	5 / 10 (Requires license server management)	Stack A
Total Cost of Ownership (TCO)	Low Capital Expenditure (Engineering cost only)	High (\$5M+ annual license renewal)	Stack A
Speed of Initial Deployment	Medium (Requires system integration)	High (Off-the-shelf software modules)	Stack B
Performance at Scale	High (Distributed Kafka/MinIO/PostGIS scales linearly)	High (Requires high hardware allocation)	Tie

Conclusion: For **BHU-NETRA, Architecture Stack A (Open-Source & Indigenous First)** is the recommended primary deployment for core Mission Control (TT&C and C2), ensuring national sovereignty and immunity from external trade/vendor restrictions.